Probability Theory - Winter term 2025/26

Personal Solution Set 3

TU Dortmund

Problem 1

Let s, t be simple functions, $E \in \mathcal{A}$ for the measurable space $(\Omega, \mathcal{A})$, $\Omega \neq \emptyset$. Let μ be a measure on $(\Omega, \mathcal{A})$. Show the following:

i) $\rho : \mathcal{A} \to [0, \infty]$ with $\rho(E) := \int_E s \, d\mu$ is a measure on $(\Omega, \mathcal{A})$.

ii) for $\alpha, \beta \in \mathbb{R}^+$ it holds

$$\int_E (\alpha s + \beta t) \, d\mu = \alpha \int_E s \, d\mu + \beta \int_E t \, d\mu.$$

Now, let $f, g : \Omega \to [0, \infty]$ be nonnegative functions. Show that

iii) for $\alpha, \beta \in \mathbb{R}^+$ it holds

$$\int_\Omega (\alpha f + \beta g) \, d\mu = \alpha \int_\Omega f \, d\mu + \beta \int_\Omega g \, d\mu.$$

Solution

i) Let $s = \sum_{i=1}^n a_i 1_{A_i}$ and $t = \sum_{j=1}^m b_j 1_{B_j}$ be simple functions where $a_i, b_j \in [0, \infty)$. $\rho(E)$ can be written as:

$$\rho(E) = \int_E s \, d\mu = \sum_{i=1}^n a_i \mu(E \cap A_i).$$

Checking properties of a measure:

- Positive mapping: μ is a measure (positive), and coefficients $a_i \geq 0$, so $\rho(E) \geq 0$.
- Empty set: $\rho(\emptyset) = \sum_{i=1}^n a_i \mu(\emptyset \cap A_i) = \sum_{i=1}^n a_i \mu(\emptyset) = 0$.
- σ-additivity: For a sequence of pairwise disjoint sets $E_1, E_2, \dots \in \mathcal{A}$:

$$\begin{aligned}
\rho\left(\bigcup_{j=1}^\infty E_j\right) &= \sum_{i=1}^n a_i \mu\left(\left(\bigcup_{j=1}^\infty E_j\right) \cap A_i\right) \\
&= \sum_{i=1}^n a_i \mu\left(\bigcup_{j=1}^\infty (E_j \cap A_i)\right) \\
&= \sum_{i=1}^n a_i \sum_{j=1}^\infty \mu(E_j \cap A_i) \\
&= \sum_{j=1}^\infty \sum_{i=1}^n a_i \mu(E_j \cap A_i) = \sum_{j=1}^\infty \rho(E_j).
\end{aligned}$$

Thus, ρ is a measure on $(\Omega, \mathcal{A})$.

ii) Assuming $A_1, \dots, A_n$ and $B_1, \dots, B_m$ form partitions of Ω, the indicator functions can be rewritten:

$$1_{A_i} = \sum_{j=1}^{m} 1_{A_i \cap B_j}, \quad 1_{B_j} = \sum_{i=1}^{n} 1_{A_i \cap B_j}.$$

The sum $s + t$ is a new simple function:

$$s + t = \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + b_j) 1_{A_i \cap B_j}.$$

Evaluating the integral:

$$\begin{aligned}
\int_E (\alpha s + \beta t)\, d\mu &= \sum_{i=1}^{n} \sum_{j=1}^{m} (\alpha a_i + \beta b_j) \mu(E \cap A_i \cap B_j) \\
&= \alpha \sum_{i=1}^{n} \sum_{j=1}^{m} a_i \mu(E \cap A_i \cap B_j) + \beta \sum_{i=1}^{n} \sum_{j=1}^{m} b_j \mu(E \cap A_i \cap B_j) \\
&= \alpha \sum_{i=1}^{n} a_i \mu \left(E \cap A_i \cap \left(\bigcup_{j=1}^{m} B_j \right) \right) + \beta \sum_{j=1}^{m} b_j \mu \left(E \cap \left(\bigcup_{i=1}^{n} A_i \right) \cap B_j \right) \\
&= \alpha \int_E s\, d\mu + \beta \int_E t\, d\mu.
\end{aligned}$$

iii) Let f, g be non-negative measurable functions. By definition of a measure integral:

$$\int_E f\, d\mu = \sup \left\{ \int_E s\, d\mu : 0 \leq s \leq f,\ s \text{ simple function} \right\}.$$

For the linear combination, knowing that combinations of simple functions are simple functions (from ii):

$$\begin{aligned}
\int_\Omega (\alpha f + \beta g)\, d\mu &= \sup \left\{ \int_\Omega (\alpha s + \beta t)\, d\mu : 0 \leq s \leq f,\ 0 \leq t \leq g \right\} \\
&= \sup \left\{ \alpha \int_\Omega s\, d\mu + \beta \int_\Omega t\, d\mu : 0 \leq s \leq f,\ 0 \leq t \leq g \right\} \\
&= \alpha \sup \left\{ \int_\Omega s\, d\mu : 0 \leq s \leq f \right\} + \beta \sup \left\{ \int_\Omega t\, d\mu : 0 \leq t \leq g \right\} \\
&= \alpha \int_\Omega f\, d\mu + \beta \int_\Omega g\, d\mu.
\end{aligned}$$

Problem 2

Let $(\mathbb{R}^+, \mathcal{B}(\mathbb{R}^+), P)$ be a probability space with $P = \text{Exp}(\lambda)$ with $\lambda > 0$, where for $0 \leq a < b$ the probability measure P is defined by

$$P((a, b]) := \exp(-\lambda a) - \exp(-\lambda b).$$

Consider the following measurable functions $X_i : (\mathbb{R}^+, \mathcal{B}(\mathbb{R}^+)) \to (\mathbb{R}^+, \mathcal{B}(\mathbb{R}^+))$ for $i = 1, 2, 3$ and show that

i) for $X_1(z) = 10z$, the image measure is $P^{X_1} = \text{Exp}\left(\frac{\lambda}{10}\right)$,

ii) for $X_2(z) = \exp(z)$, the image measure is

$$P^{X_2}((a,b]) = \begin{cases} \left(\frac{1}{a}\right)^\lambda - \left(\frac{1}{b}\right)^\lambda, & \text{if } a \geq 1, \\ 1 - \left(\frac{1}{b}\right)^\lambda, & \text{if } a < 1, b \geq 1, \\ 0, & \text{if } b < 1, \end{cases}$$

iii) for $X_3(z) = 10$, the image measure is $P^{X_3} = \delta_{10}$.

Hint: Given measurable spaces $(\Omega_1, \mathcal{A}_1)$ and $(\Omega_2, \mathcal{A}_2)$, a measurable mapping $X : (\Omega_1, \mathcal{A}_1) \to (\Omega_2, \mathcal{A}_2)$ and a measure $\mu : \mathcal{A}_1 \to [0, \infty]$, the image measure μ^X of μ is defined as $\mu^X(A) = \mu(X^{-1}(A))$, for $A \in \mathcal{A}_2$.

Solution

Using the definition of the image measure $P^X((a,b]) = P(X^{-1}((a,b]))$:

i) For $X_1(z) = 10z$:

$$\begin{aligned} P^{X_1}((a,b]) = P(X_1^{-1}((a,b])) &= P\left(\left(\frac{a}{10}, \frac{b}{10}\right]\right) \\ &= \exp\left(-\lambda\frac{a}{10}\right) - \exp\left(-\lambda\frac{b}{10}\right) \\ &= \exp\left(-\frac{\lambda}{10}a\right) - \exp\left(-\frac{\lambda}{10}b\right). \end{aligned}$$

Comparing to the definition of P, this matches an exponential distribution with parameter $\frac{\lambda}{10}$, so $P^{X_1} = \mathrm{Exp}\left(\frac{\lambda}{10}\right)$.

ii) For $X_2(z) = \exp(z)$:
Case 1: $a \geq 1$. Since $z \geq 0 \Rightarrow X_2(z) \geq 1$, $\log(a) \geq 0$:

$$\begin{aligned} P^{X_2}((a,b]) &= P(X_2^{-1}((a,b])) = P((\log(a), \log(b)]) \\ &= \exp(-\lambda\log(a)) - \exp(-\lambda\log(b)) \\ &= a^{-\lambda} - b^{-\lambda} = \left(\frac{1}{a}\right)^\lambda - \left(\frac{1}{b}\right)^\lambda. \end{aligned}$$

Case 2: $a < 1$ and $b \geq 1$. Here $\log(a) < 0$, but the domain is $\mathbb{R}^+$, so the pre-image intersects with $\mathbb{R}^+$ starting from 0:

$$\begin{aligned} P^{X_2}((a,b]) &= P((0, \log(b)]) \\ &= \exp(-\lambda \cdot 0) - \exp(-\lambda\log(b)) = 1 - \left(\frac{1}{b}\right)^\lambda. \end{aligned}$$

Case 3: $b < 1$. Here $\log(b) < 0$. The pre-image has no intersection with $\mathbb{R}^+$:

$$P^{X_2}((a,b]) = P(\emptyset) = 0.$$

iii) For $X_3(z) = 10$: The pre-image depends solely on whether 10 is inside the interval $(a,b]$:

$$X_3^{-1}((a,b]) = \{y \in \mathbb{R}_+ : 10 \in (a,b]\} = \begin{cases} \Omega, & 10 \in (a,b] \\ \emptyset, & 10 \notin (a,b] \end{cases}$$

Calculating the image measure:

$$P^{X_3}((a,b]) = P(X_3^{-1}((a,b])) = \begin{cases} P(\Omega) = 1, & 10 \in (a,b] \\ P(\emptyset) = 0, & 10 \notin (a,b] \end{cases}$$

This defines the Dirac measure at 10, so $P^{X_3} = \delta_{10}$.

Problem 3

1. Consider the measure space $([a,b], \mathcal{B}([a,b]), \lambda)$ with $\mathcal{B}([a,b])$ the Borel-σ-field on $[a,b]$ and the Lebesgue measure. Show that any singleton set $\{c\}$, $c \in [a,b]$ is Lebesgue-measurable and determine $\lambda(\{c\})$.
2. Prove that $\mathbb{Q} \cap [a,b]$ is Lebesgue measurable and calculate $\lambda(\mathbb{Q} \cap [a,b])$.
3. What are the consequences of the first two items for the Lebesgue measures of the intervals $\lambda((a,b))$, $\lambda((a,b])$, $\lambda([a,b))$?

Hint: Every measure μ is continuous from above, i.e.: $A_{n+1} \subseteq A_n \; \forall n \in \mathbb{N}$ implies $\mu(A_n) \longrightarrow_{n\to\infty} \mu(\bigcap_{k=1}^{\infty} A_k)$.

Solution

1. Defining a decreasing sequence $A_n := [c - \frac{1}{n}, c + \frac{1}{n}]$. These closed intervals are Lebesgue-measurable as they generate the Borel-σ-field. Taking the intersection:
$$\bigcap_{n=1}^{\infty} A_n = \{c\}.$$
Using continuity from above (from the hint):
$$\begin{aligned}\lambda(\{c\}) = \lambda\left(\bigcap_{n=1}^{\infty} A_n\right) &= \lim_{n\to\infty} \lambda\left(\left[c - \frac{1}{n}, c + \frac{1}{n}\right]\right) \\ &= \lim_{n\to\infty}\left(c + \frac{1}{n} - \left(c - \frac{1}{n}\right)\right) = \lim_{n\to\infty} \frac{2}{n} = 0.\end{aligned}$$
The singleton set $\{c\}$ is in the Borel-σ-field, so it is Lebesgue-measurable.
2. Representing the rational numbers in the interval as a countable union of disjoint singleton sets:
$$\mathbb{Q} \cap [a,b] = \bigcup_{q \in \mathbb{Q} \cap [a,b]} \{q\}.$$
Since it is a countable union of Borel sets, it is Lebesgue-measurable. Because the Lebesgue measure is σ-additive:
$$\lambda(\mathbb{Q} \cap [a,b]) = \sum_{q \in \mathbb{Q} \cap [a,b]} \lambda(\{q\}) = 0.$$
3. The intervals can be expressed as:
$$\begin{aligned}[a,b) &= [a,b] \setminus \{b\} \\ (a,b] &= [a,b] \setminus \{a\} \\ (a,b) &= [a,b] \setminus (\{a\} \cup \{b\})\end{aligned}$$
Since the measure of singleton sets is zero (from part 1), subtracting them does not change the total measure. Therefore:
$$\begin{aligned}\lambda([a,b)) &= \lambda([a,b]) - \lambda(\{b\}) = \lambda([a,b]) \\ \lambda((a,b]) &= \lambda([a,b]) - \lambda(\{a\}) = \lambda([a,b]) \\ \lambda((a,b)) &= \lambda([a,b]) - \lambda(\{a\}) - \lambda(\{b\}) = \lambda([a,b]).\end{aligned}$$
All these intervals are mapped to the same value by the Lebesgue measure.